# Characterization of Human Balance Control During Very Small Sensory Conflict Velocities

**Master's thesis**

submitted by

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## Abstract

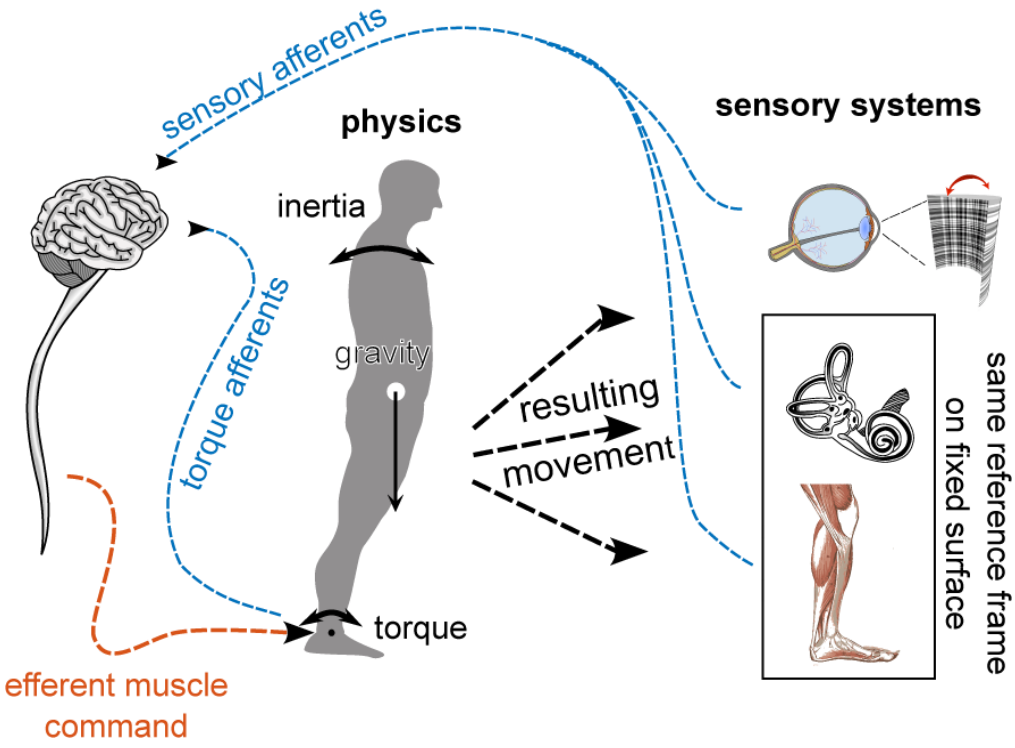
Maintaining upright balance requires the central nervous system to perform multisensory integration, combining information from the visual, vestibular, and proprioceptive systems. Previous studies have shown that postural response gain decreases as visual scene velocity increases. However, extending this relationship toward zero velocity would predict an unbounded increase in gain. A recently proposed model of sensory integration, the Reference Frame Motion (RFM) estimator, instead predicts that the effective visual contribution forms a plateau below a certain velocity of the visual surround in space. Given that such low visual scene velocities have not been investigated previously, this study examined whether postural response gain approaches a plateau in this range. Twelve healthy participants stood upright with a virtual reality (VR) head mounted display and were positioned in front of a high contrast, half cylindrical visual screen that moved in the anterior posterior direction at constant velocities ranging from 0.01 to 10°/s. Body motion was recorded using two motion trackers, and the whole body center of mass (COM) movement was derived thereof. The ratio between COM movement and visual stimulus amplitude was calculated at the fundamental stimulus frequency of 0.5 Hz. Gain decreased with increasing visual scene velocity. Between 0.03 and 0.25°/s, gain values were relatively similar, suggesting a tendency toward levelling off at low velocities. However, the 0.01°/s condition showed a higher mean gain and substantially greater variability, preventing the clear identification of a plateau. The velocity range in which gain appeared to level off was close to the dead zone threshold of approximately 0.28°/s estimated from the RFM model fit. These findings suggest that gain may no longer continue to increase as visual scene velocity approaches zero. This pattern supports the proposed RFM mechanism, including a low velocity plateau. Further measurements at very low velocities are needed to determine this relationship more accurately.

## Background

Maintaining an upright posture requires the central nervous system (CNS) to continuously monitor body orientation and correct emerging postural deviations. As gravitational forces tend to amplify these deviations, corrective torque must be generated using sensory information to prevent instability (Peterka, 2002). These corrections rely on multisensory integration (MSI) and form part of a closed loop system in which sensory feedback, motor activation, and body mechanics influence one another (Peterka et al., 2018). This process depends mainly on the sensory information from the visual, vestibular, and proprioceptive systems which contribute to postural control (Assländer & Peterka, 2014), as illustrated in Figure 1.

**Figure 1**

*Schematic representation of the closed loop postural control system. Reproduced from Assländer et al. (2025).*



A challenge of MSI is that conflicting sensory inputs can provide different interpretations of the same event. Relative motion between the body and the visual environment may be caused by self motion, environmental motion, or both, making the source of changes in visual sensory signals ambiguous (Dokka et al., 2010). Assländer et al. (2025) illustrated this ambiguity using the example of a moving train. This example highlights the CNS's ability to resolve sensory conflicts. Similar sensory conflicts have been reproduced experimentally by moving the visual scene during upright standing (Peterka & Benolken, 1995; Peterka, 2002; Mergner et al., 2005; Assländer et al., 2025). The resulting body sway responses provide a measurable indication of how visual information influences postural control.

Peterka (2002) showed that, within a fixed sensory condition and stimulus amplitude, the relationship between visual scene motion and body sway was quasi linear. However, comparisons across stimulus amplitudes revealed nonlinear behaviour in response gain, which represents the ratio of body sway response amplitude to visual scene stimulus amplitude at the corresponding frequency (Peterka, 2002), with gain declining as the sway response approached saturation (Mergner et al., 2005). This nonlinear response behaviour has traditionally been attributed to sensory reweighting, an active adjustment of the relative contributions of visual, vestibular, and proprioceptive information through the CNS (Peterka, 2002). Assländer and Peterka (2014) further demonstrated that sensory reweighting is a dynamic process, with sensory contributions adjusting over time following changes in stimulus amplitude. Although sensory weights quantify changes in sensory contributions, they do not reveal the mechanism through which the CNS produces these changes (Assländer et al., 2025).

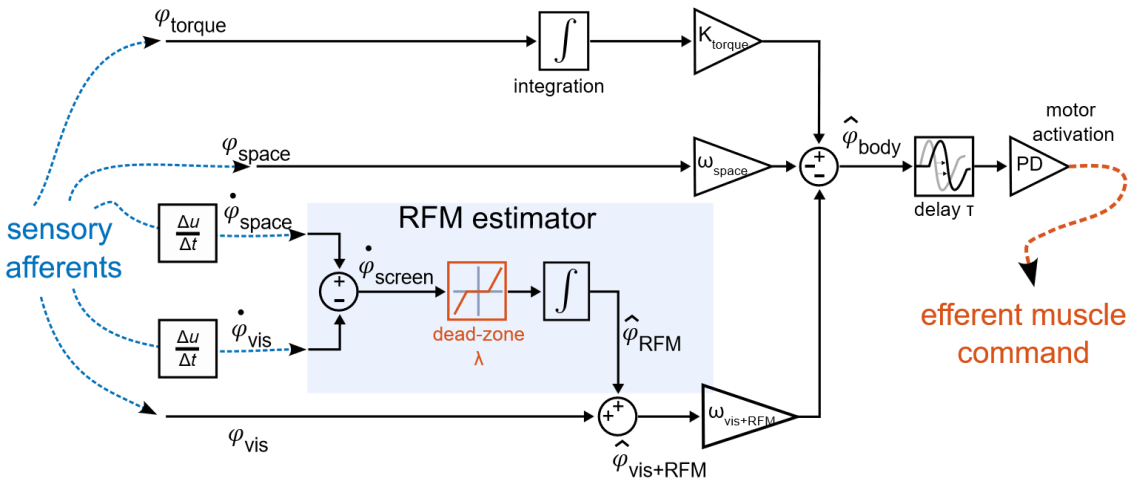
To account for the nonlinear effect of visual motion on postural responses, Dokka et al. (2010) proposed a Bayesian model describing how the CNS may resolve this ambiguity. The model reproduced a power law relationship between visual scene velocity and postural

response gain, with gain decreasing as visual scene velocity increased on loglog scales. However, directly extrapolating this relationship towards zero velocity would predict an infinite gain, indicating that the power law approximation cannot explain responses at very low visual scene velocities.

An alternative mechanism for resolving sensory conflicts involves estimating the motion of sensory reference frames (Assländer et al., 2025). This reconstruction mechanism was first proposed by Mergner et al. (1991) and was later incorporated into a broader framework for balance control (Mergner et al., 2003; Mergner, 2010; Assländer et al., 2015). Building on this framework, Assländer et al. (2025) used a reductionist approach to develop the Reference Frame Motion (RFM) estimator (Figure 2), which explains changes in sensory contributions without explicitly adjusting sensory weights.

**Figure 2**

*Schematic representation of the Reference Frame Motion (RFM) estimator and model parameters estimated from experimental data. The nonlinear dead zone threshold is denoted by λ and was estimated at 0.28°/s. Reproduced from Assländer et al. (2025).*




| P / mgh (1/rad) | D / mgh (s/rad) | $K_{torque}$ (rad/ Nms) | τ (s) | $\omega_{vis+RFM}$ | λ (° / s) | mgh (Nm) | J (kg m²) |
|---|---|---|---|---|---|---|---|
| 1.22 | 0.45 | $6.3 * 10^{-5}$ | 0.20 | 0.264 | 0.28 | 680 | 79.3 |

The RFM estimator reconstructs reference frame motion and uses this estimate to correct the corresponding sensory input. A crucial element of the estimator is a nonlinear dead zone that prevents correction when the estimated visual scene velocity remains below a certain velocity threshold (Assländer et al., 2025). Once the threshold is exceeded, the estimated reference frame motion reduces the effective visual contribution. Thus, changes in sensory contributions result from correcting reference frame motion rather than adjusting sensory weights.

Assländer et al. (2025) demonstrated that the RFM estimator accounts for the nonlinear relationship between visual stimulus amplitude and body sway amplitude, in which sway increased less than proportionally as the stimulus became larger. The estimator also predicted spectral distortions, characterized by additional frequency components in the sway response that were absent from the visual stimulus. These predictions were confirmed experimentally. Across the seven visual stimulus conditions, the model accounted for 81–95% of the variance in the measured sway responses. Furthermore, the estimator reproduced the previously reported power law relationship between visual scene velocity and postural response gain over a broad range of velocities.

The RFM estimator predicts that the effective visual contribution forms a plateau while visual scene velocity remains below the dead zone threshold and decreases hyperbolically once this threshold is exceeded (Assländer et al., 2025). Previous findings from Peterka and Benolken (1995) and Mergner et al. (2005) suggest that response gain may deviate from this decline at low stimulus velocities, indicating the possible emergence of such a plateau. In contrast, extending the power law relationship described by Dokka et al. (2010) toward zero velocity would predict a continuous increase in response gain. However, this power law approximation is only expected to hold when visual scene velocity is sufficiently large and may therefore not accurately describe responses at very low visual scene velocities. Despite

these findings, postural responses in the very low velocity range have not yet been investigated systematically. This leaves an important gap in understanding how visual contributions to balance control behave as stimulus velocity approaches zero. Characterizing this range could therefore provide important information for evaluating different models of sensory integration and for better understanding the mechanisms underlying sensory conflict resolution.

Previous studies have commonly investigated visually induced postural sway using sine wave visual motion (Peterka & Benolken, 1995; Mergner et al., 2005; Assländer et al., 2025). However, several studies suggest that changes in sensory contributions depend on stimulus velocity. Sine wave stimuli have continuously changing velocities and may cause dynamic changes in sensory contributions, leading to distortions in the sway response (Assländer et al., 2025). Therefore, we used ramps which have a fixed velocity and only inverted directions, to avoid distortions. The primary aim was to determine whether postural response gain levels off at very low visual scene velocities. The identification of such a transition would be consistent with the dead zone threshold predicted by the RFM estimator and would provide a finite description of gain near zero velocity, where extrapolation of the power law relationship predicts an unbounded increase.

## Methods

The experiment consisted of two sessions and thirteen participants were recruited. One participant did not attend the second session and was therefore excluded from the analysis. This resulted in a sample of 12 participants (7 female, 5 male; age: 25.6 ± 2.7 years, range: 22–32 years). Participants reported no history of balance disorders. Anthropometric characteristics included a mean body height of 1.67 ± 0.14 m and a mean body mass of 67.8 ± 17.7 kg. Participants provided written informed consent and received €15 for each completed

experimental session. The experimental protocol was conducted in accordance with the regulations of the Institutional Review Board of the University of Konstanz and in accordance with the Declaration of Helsinki.

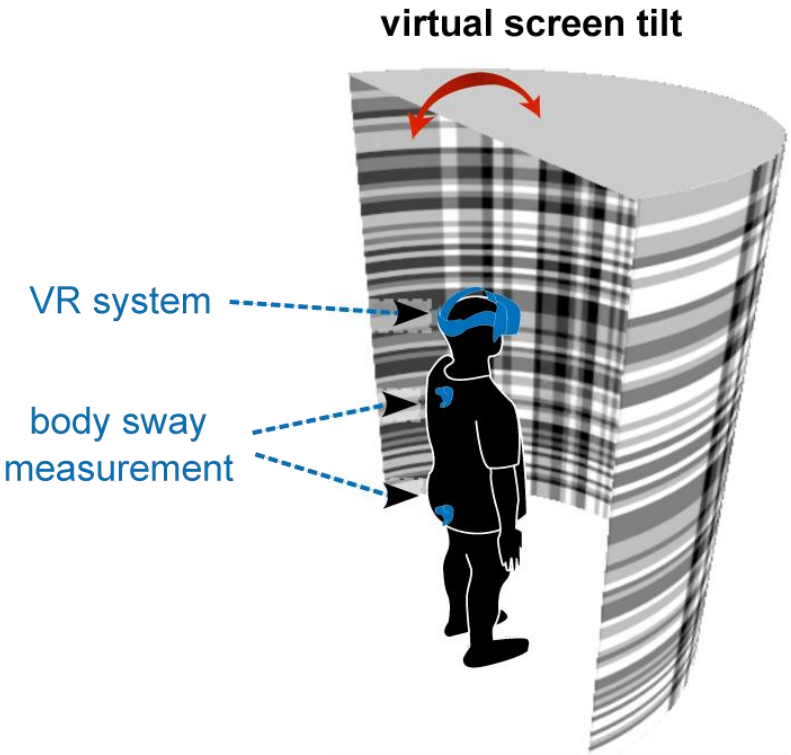
Participants stood upright with a virtual reality (VR) head mounted display (Vive Pro Eye, HTC, Taoyuan, Taiwan). The experimental setup is illustrated in Figure 3**.** Body motion was recorded using two motion trackers (Vive Tracker 3.0, HTC, Taoyuan, Taiwan). One tracker was positioned between the shoulder blades and attached to a harness worn by the participant. The second tracker was secured at the height of the greater trochanter using a Velcro strap. Force plate data were additionally recorded using a force plate but were not included in the analysis. Within the virtual environment, participants were positioned in front of a high contrast, half cylindrical visual screen with a radius of 1 m. The screen rotated in the anterior posterior direction. Its rotational axis was located approximately at the level of the ankle joints, 8.8 cm above the floor. Tracker positions were recorded at approximately 90 Hz with every display update. Custom software developed in Unity (Unity Technologies) and SteamVR (Valve) was used to generate the virtual environment, present the stimuli, and record motion tracking and analogue data. Previous validation showed that this setup evokes balance behaviour comparable to that observed in real world moving visual environments (Assländer et al., 2023). SteamVR motion smoothing and automatic resolution adjustment were enabled to maintain a stable frame rate throughout the experiment.

Participants were instructed to stand barefoot in an upright and comfortable position, with their hands resting one over the other in front of the body. Each session included 12 trials. The first session included 11 trials lasting 210s and one 300s pseudorandom ternary sequence (PRTS) trial. The second session included 10 trials lasting 210s and two 300s PRTS trials. The visual stimuli consisted of repeated triangular screen waveforms with a cycle duration of 2s. The screen rotated at a constant angular velocity for 1s before reversing direction for the

following 1s. Each 210s trial therefore contained 105 complete cycles. The peak screen angle was determined by the stimulus velocity and corresponded to half its numerical value in degrees. During the first session, one trial was performed without visual scene motion, two trials at 0.01°/s, two trials at 0.05°/s, and one trial at each of the following velocities: 0.15, 0.25, 0.5, 1, 5, and 10°/s. During the second session, one trial was performed without visual scene stimuli, one trial at 0.01°/s, three trials at 0.03°/s, two trials at 0.07°/s, and one trial at each of the following velocities: 0.1, 0.2, and 0.3°/s. The PRTS trials and force plate recordings were not included in the analyses reported in the present study.

**Figure 3**

*Schematic representation of the experimental setup, showing the virtual screen and the positions of the body trackers. Reproduced from Assländer et al. (2025).*

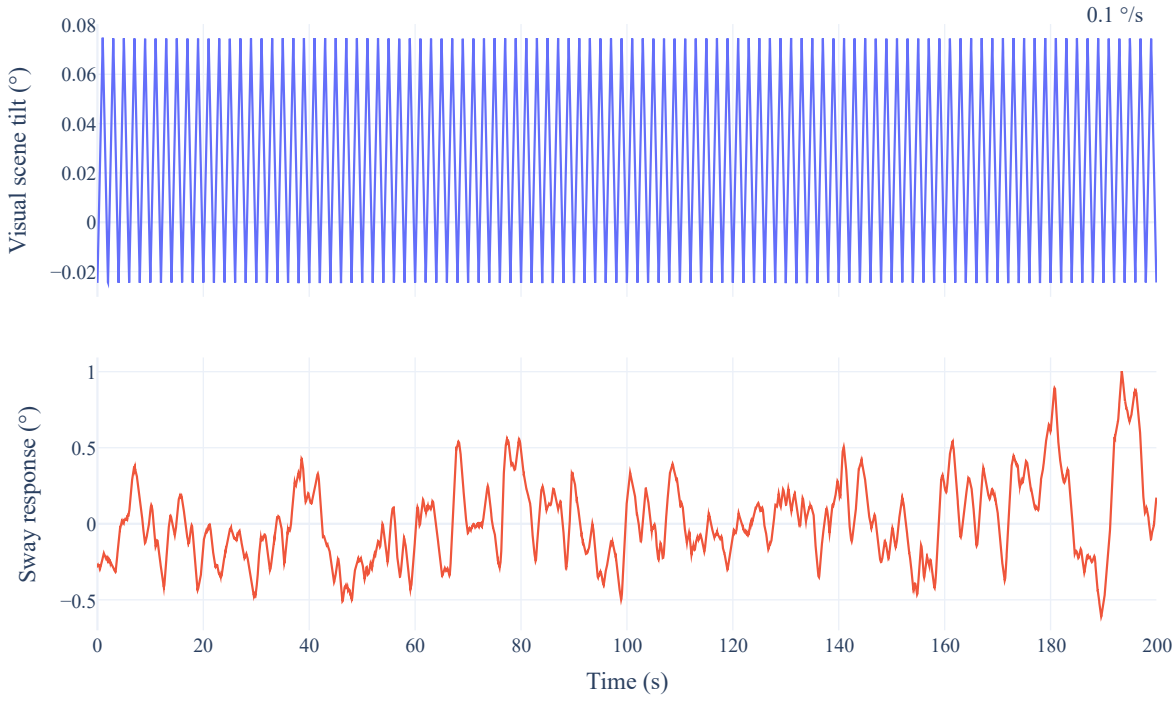


## Data Analysis

The whole body centre of mass (COM) angular displacement in the anterior posterior direction was estimated from the shoulder and hip tracker trajectories and participant height using a two segment anthropometric model (Assländer et al., 2023). The visual stimulus and COM signals were resampled to 90 Hz using cubic interpolation, and no additional temporal filtering was applied. Approximately the first 10s of each trial were discarded to exclude initial transient responses. An example 200s segment of the visual scene tilt and corresponding sway response signals used for the analysis is shown in Figure 4.

**Figure 4**

*Representative time domain signals from the 0.1°/s visual scene velocity condition. The upper panel shows the visual scene tilt, and the lower panel shows the corresponding sway response during the 200s segment. The visual scene stimulus followed a triangular waveform with a cycle duration of 2s, corresponding to a fundamental frequency of 0.5 Hz.*



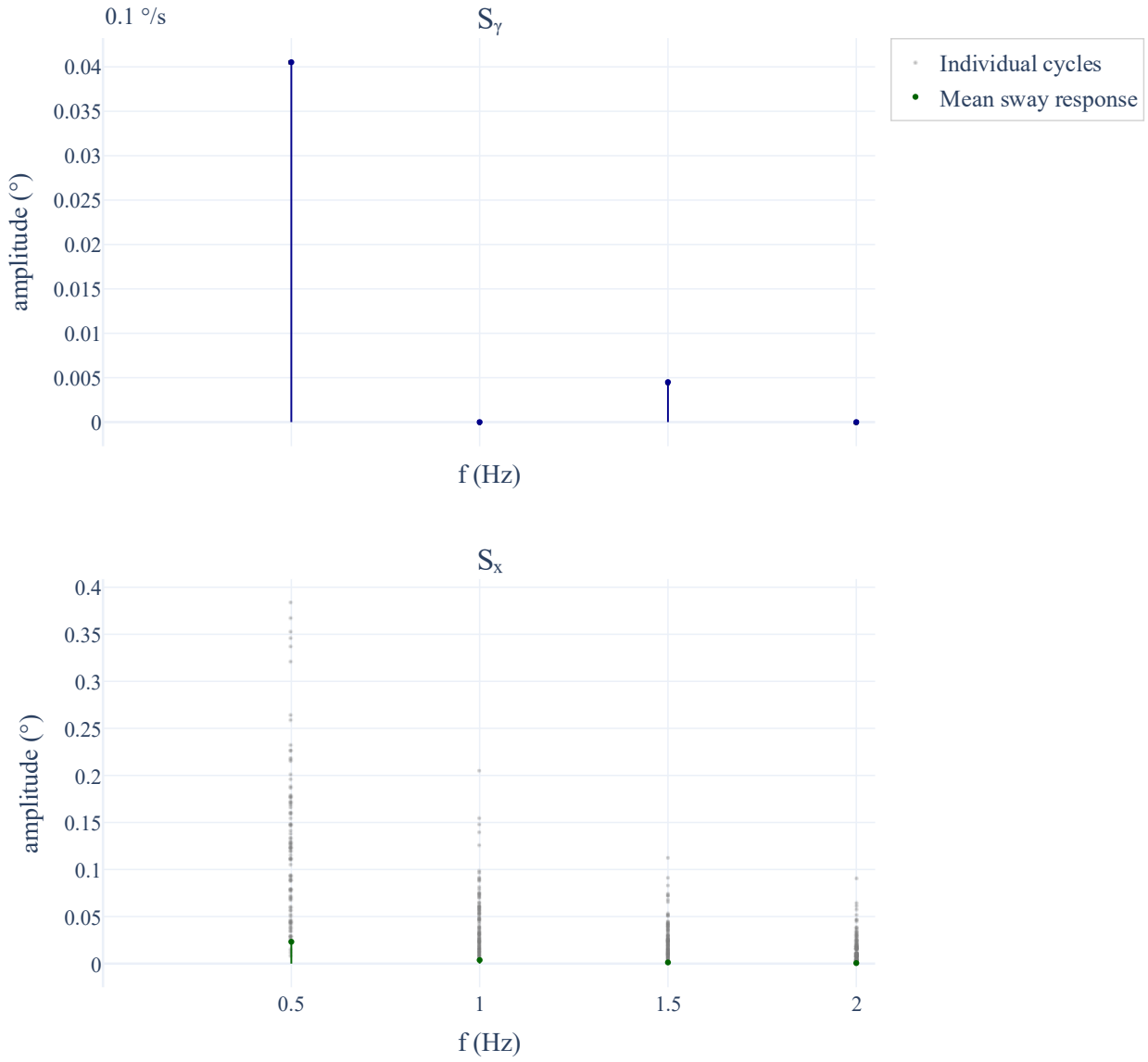
The subsequent 200s were divided into 100 cycles of 2s, with each cycle containing 180 samples, and the remaining incomplete segment was excluded. Trials containing missing files or identified as invalid during quality screening were excluded from further analysis. The visual scene stimulus was denoted as y(t), and the sway response as x(t). Their corresponding amplitude spectra were denoted as $S_y$ and $S_x$, respectively. Fourier spectra were calculated separately for each stimulus and response cycle. For the descriptive stimulus and sway response figures, spectral estimates were averaged across cycles, repeated trials, and participants for each stimulus condition. For the gain analysis, the frequency response function (FRF) was calculated separately for each participant and stimulus condition by dividing the averaged cross power spectrum between the sway response and visual stimulus by the averaged stimulus power spectrum, with averaging performed across cycles and repeated trials. Gain and phase were obtained from the resulting participant specific FRFs. Participant gain values were then summarized using the arithmetic mean and 95% t-based confidence intervals.

## Results

The amplitude spectra from one trial of one participant at the 0.1°/s velocity are shown in Figure 5. As expected from the 2s triangular waveform, the stimulus spectrum showed a dominant component at 0.5 Hz and a smaller component at 1.5 Hz. The 0.5 Hz component was also clearly visible in the mean sway response spectrum. The individual cycle sway spectra showed considerable variability in amplitude across frequencies, whereas averaging across cycles produced a smoother spectrum in which the main frequency components were more clearly distinguishable. The mean sway response spectrum showed a substantial component at 0.5 Hz and smaller mean amplitudes at 1, 1.5, and 2 Hz, with considerable variability at the higher frequency points.

**Figure 5**

*Stimulus and sway response amplitude spectra for the 0.1°/s condition from one trial of one participant. The upper panel shows the visual scene stimulus spectrum. The lower panel shows the individual cycle sway response spectra in grey and the mean sway response spectrum in green, obtained by averaging the spectra across cycles before taking the absolute value.*



The pooled mean time domain signals for the no stimulus condition and visual scene velocities of 0.01, 0.1, 1, and 10°/s are shown in Figure 6. The visual scene angle illustrated the triangular waveform across all stimulus conditions, with its amplitude increasing at higher stimulus velocities. The corresponding sway responses showed an oscillatory pattern that became more clearly visible as stimulus velocity increased. However, the increase in sway amplitude was considerably smaller than the increase in visual scene amplitude, particularly at the higher velocities. Between 0.1 and 1°/s, the visual scene amplitude increased by a factor of 10, whereas the sway amplitude increased only from approximately 0.05° to 0.08°. At 10°/s, the sway amplitude remained approximately the same.

Figure 7 shows the corresponding pooled mean amplitude spectra for the selected visual scene velocities. Across conditions, the largest sway response component was observed at the fundamental stimulus frequency of 0.5 Hz. At this frequency, the sway response amplitude increased from approximately 0.002° at 0.01°/s to 0.02° at 0.1°/s and approximately 0.03° at 1°/s, with no additional increase at 10°/s. Thus, although the stimulus amplitude increased proportionally across the selected visual scene velocities, the sway response amplitude increased less than proportionally at the higher velocities.

**Figure 6**

*Pooled mean visual scene stimulus and sway response signals in the time domain across the stimulus conditions shown. The left column shows the mean visual scene stimulus, y(t), and the right column shows the mean sway response, x(t). Signals were pooled across all included cycles, repeated trials, and participants. Rows correspond to no stimulus and visual scene velocities of 0.01, 0.1, 1, and 10°/s.*

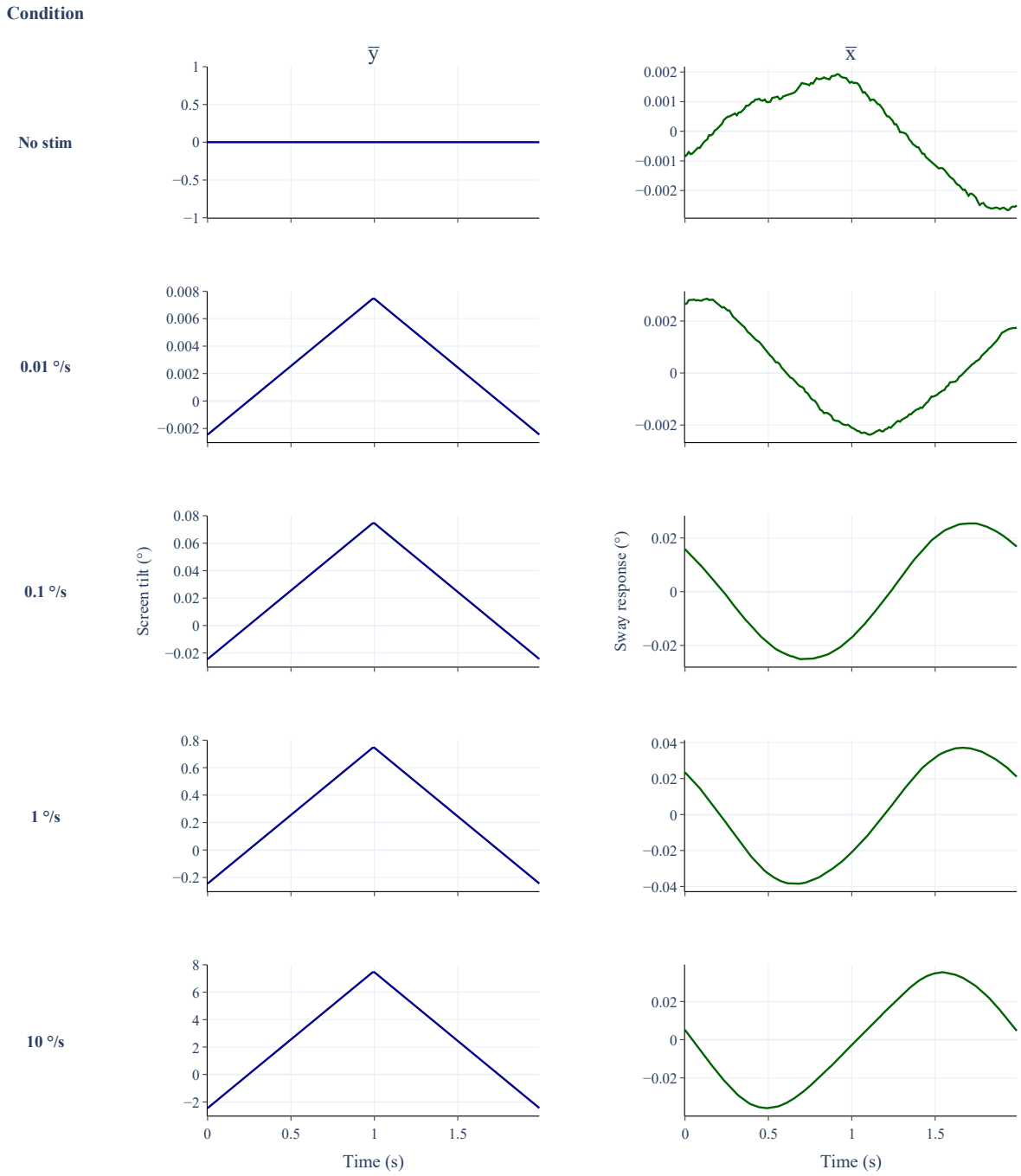
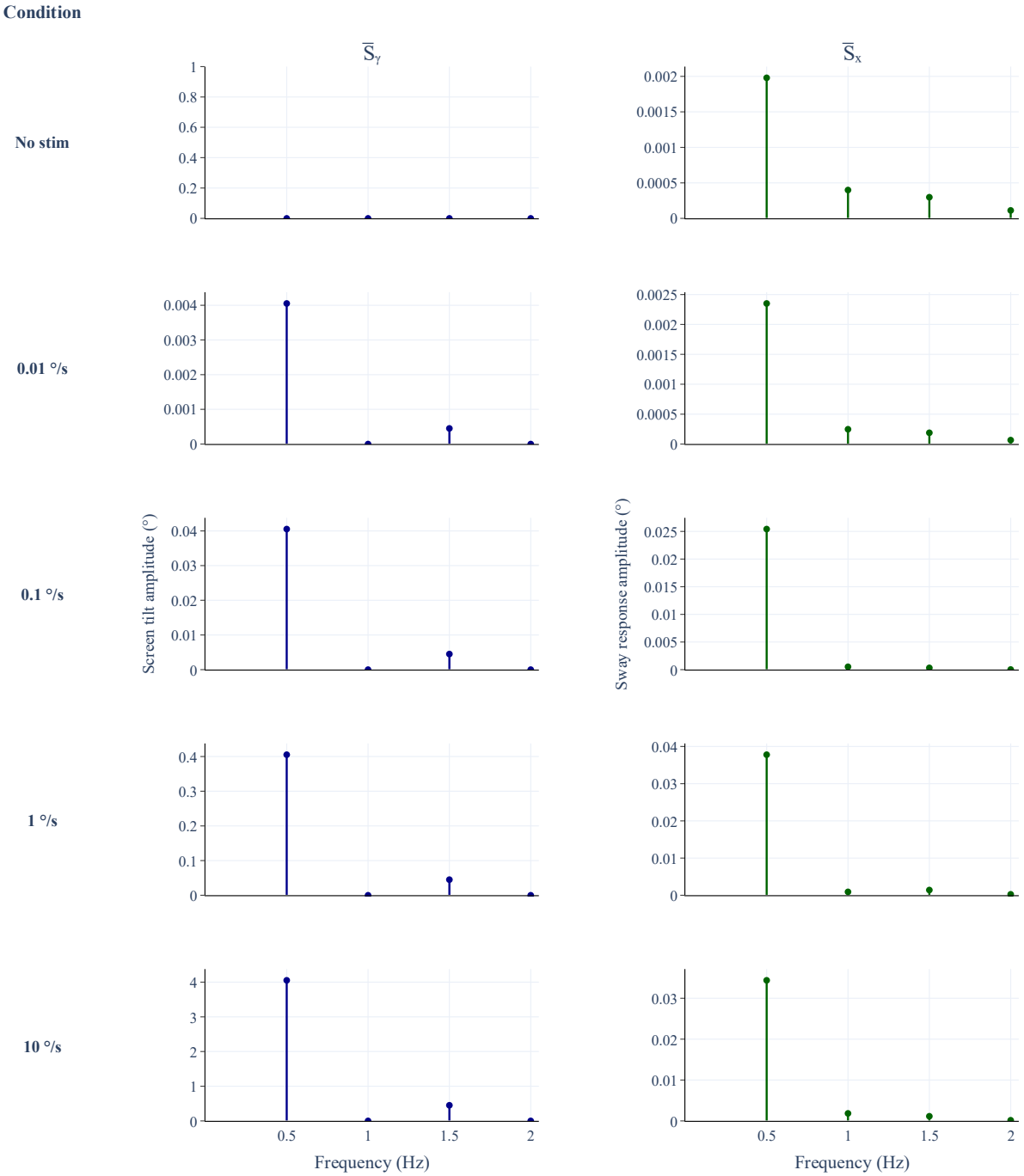


**Figure 7**

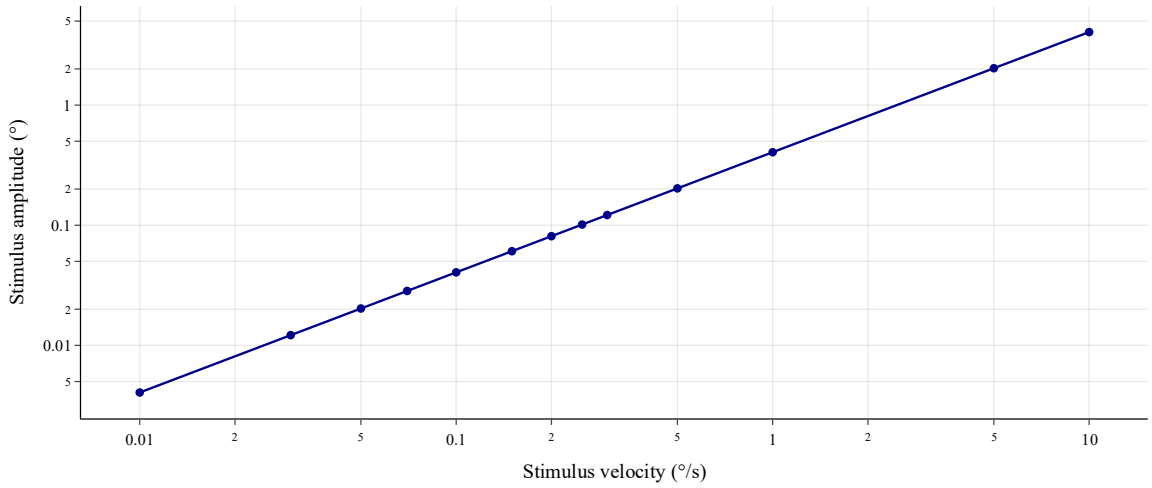
*Pooled mean visual scene stimulus and sway response amplitude spectra across the stimulus conditions shown. The left column shows the mean visual scene stimulus spectrum, $S_y$, and the right column shows the mean sway response spectrum, $S_x$. Spectra were pooled across all included cycles, repeated trials, and participants. Rows correspond to no stimulus and visual scene velocities of 0.01, 0.1, 1, and 10°/s.*



As illustrated in Figure 8, the relationship between stimulus amplitude and visual scene velocity is demonstrated for all conditions that were tested in both experimental sessions. The proportional increase in stimulus amplitude with visual scene velocity was maintained across the full range of stimulus conditions.

**Figure 8**

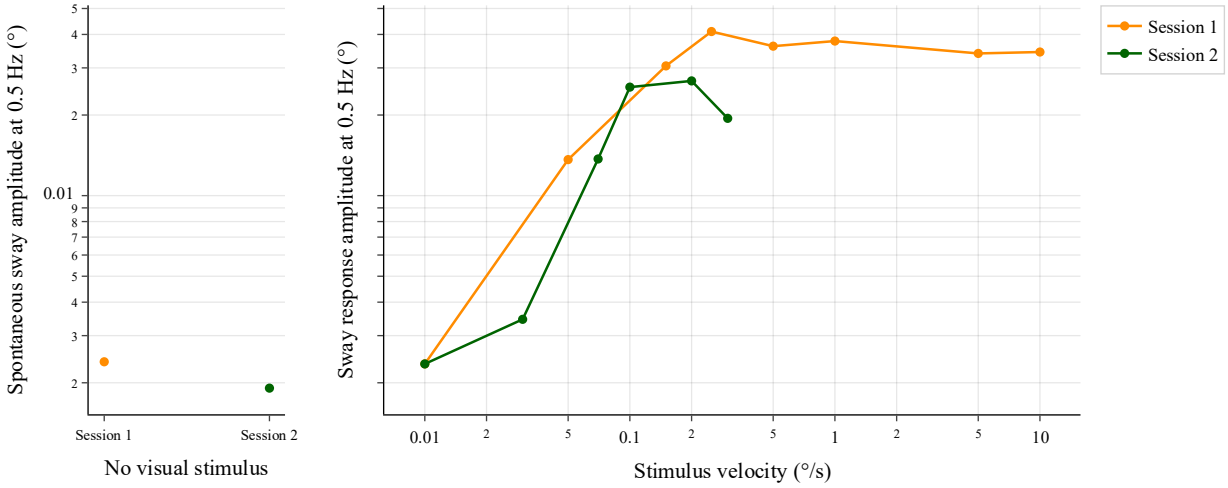
*Visual stimulus amplitude at 0.5 Hz as a function of visual scene velocity across all stimulus conditions.*




In the absence of visual stimulation, spontaneous sway amplitude at 0.5 Hz was approximately 0.0023° in Session 1 and 0.0019° in Session 2 (Figure 9). The sway response amplitude increased with visual scene velocity over the lower stimulus velocity range (Figure 9). In Session 1, the response increased from approximately 0.0023° at 0.01°/s to around 0.04° at 0.25°/s, after which relatively little additional increase was observed up to 10°/s. Session 2 provided a more detailed representation of the low velocity range, where the mean response showed minimal change between 0.01 and 0.03°/s, increased from approximately 0.006° at 0.03°/s to 0.03° at 0.2°/s, and decreased to approximately 0.023° at 0.3°/s.

**Figure 9**

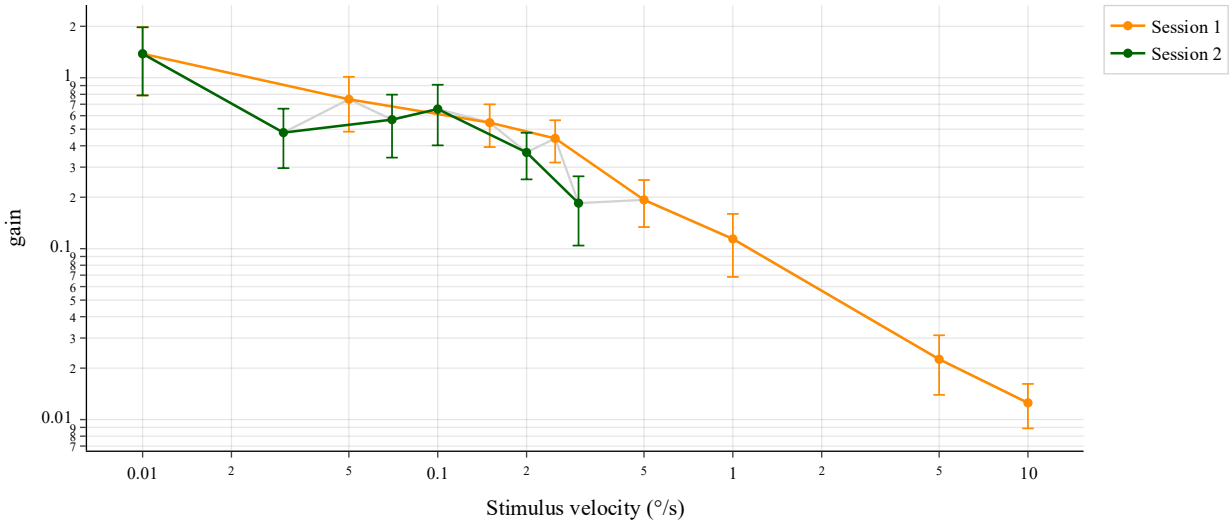
*Sway amplitude at 0.5 Hz during visual stimulation and no stimulus conditions, shown separately for Sessions 1 and 2. The left panel shows sway amplitude without visual stimulation, and the right panel shows pooled sway response amplitude as a function of visual scene velocity. For both panels, spectra were averaged across all included cycles and trials before taking the magnitude at 0.5 Hz.*




As demonstrated in Figure 10, participant gain was highest at 0.01°/s, reaching approximately 1.4, with the widest 95% confidence interval observed at this condition (0.78–1.97). The 0.01°/s condition was clearly separated from the neighbouring low velocity conditions by its higher mean gain and larger variability. Between velocities from 0.03 to 0.25°/s, gain values were relatively close in magnitude, ranging from approximately 0.3 to 0.7. At 0.03°/s, gain was approximately 0.48 (95% CI: 0.29–0.66), followed by approximately 0.57 (95% CI: 0.34–0.8) at 0.07°/s, 0.65 (95% CI: 0.40–0.91) at 0.1°/s, 0.54 (95% CI: 0.39–0.7) at 0.15°/s, and 0.44 (95% CI: 0.32–0.56) at 0.25°/s. At higher velocities, gain declined to approximately 0.19 at 0.5°/s, 0.11 at 1°/s, 0.02 at 5°/s, and 0.01 at 10°/s.

**Figure 10**

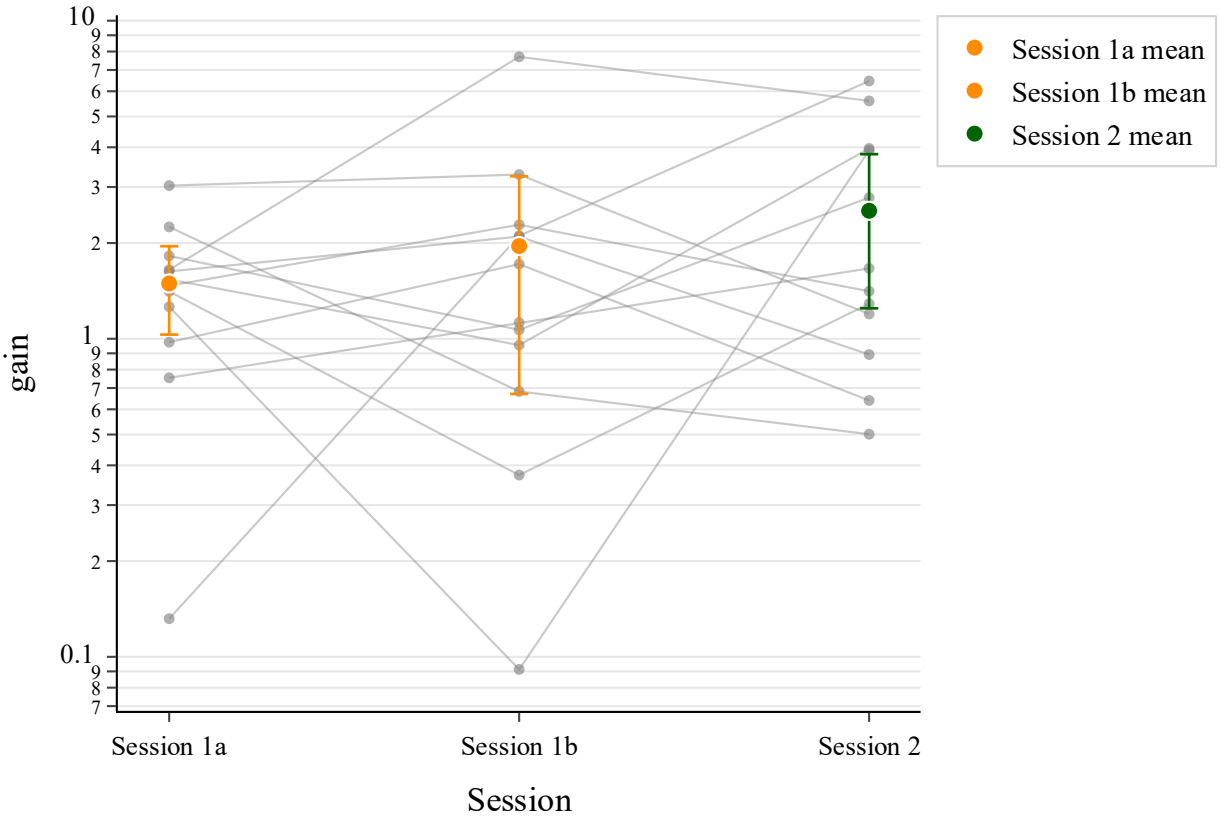
*Participant gain at the fundamental stimulus frequency (0.5 Hz) as a function of visual scene velocity. Points represent the arithmetic mean of participant gain, error bars indicate 95% t-based confidence intervals. The grey line connects the mean gain values across all stimulus velocities.*




In order to examine the repeated measurements at 0.01°/s, gain was analysed separately for the two trials in Session 1 and the trial in Session 2 (Figure 11). The mean gain was approximately 1.5 (95% CI: 1.03–1.95) in Session 1a, 1.9 (95% CI: 0.67–3.24) in Session 1b, and 2.5 (95% CI: 1.24–3.80) in Session 2. Participant values showed considerable variability across all three trials, with the largest spread observed in Session 1b. At the individual level, gain varied between the three measurements, with increases and decreases occurring across consecutive trials.

**Figure 11**

*Participant gain at 0.01°/s for the two trials in Session 1 (Session 1a and Session 1b) and the trial in Session 2. Lines connect values from the same participant across the three measurements. Colored points and error bars indicate the group mean and 95% confidence interval for each trial. Gain was evaluated at the fundamental stimulus frequency of 0.5 Hz.*




## Discussion

The present study investigated sway responses to extremely small visual scene velocities. The primary objective of this analysis is to determine whether response gain plateaus at very low velocities. Gain values between 0.03 and 0.25°/s showed considerable similarity, while the 0.01°/s condition showed a higher mean gain and substantially greater variability. Although the present results do not provide conclusive evidence for a plateau at low visual scene velocities, the relative similarity of the gain values between 0.03 and 0.25°/s suggests

that response gain may begin to level off as visual scene velocity approaches zero. At higher velocities, gain decreased with increasing visual scene velocity, consistent with previous literature (Mergner et al., 2005; Dokka et al., 2010; Assländer et al., 2025).

The observed tendency towards leveling off at low visual scene velocities is consistent with the prediction of the RFM estimator. In this model, slow reference frame motion below the dead zone threshold is not corrected. Once the threshold is exceeded, the effective visual contribution is progressively reduced (Assländer et al., 2025). In the model fit reported by Assländer et al. (2025), the estimated dead zone threshold was approximately 0.28°/s (Figure 2). The relatively similar gain values observed between 0.03 and 0.25°/s thus lie approximately within the velocity range below this estimated threshold. However, the higher and more variable gain at 0.01°/s prevents the present data from showing a clear plateau and does not allow a precise threshold to be identified. Related threshold behavior has been described in previous work. Blümle et al. (2006) found a threshold effect related to visual scene velocity and showed that visual motion began to modulate postural responses at approximately 0.31°/s, with stronger effects observed as visual scene velocity increased. However, the authors associated this effect with conscious detection of visual scene motion, suggesting a mechanism distinct from the dead zone mechanism proposed by the RFM estimator.

These findings are also relevant when considered in relation to the power law relationship described by Dokka et al. (2010). According to this model, response gain increases as visual scene velocity decreases. However, if this relationship continues toward zero velocity, gain would continue to increase towards infinity. Dokka et al. argued that this behaviour would not be plausible and suggested that a different strategy may occur at very small visual scene velocities. In contrast, the current results indicated a potential tendency for the gain to level

off at the low velocity range, thus supporting the hypothesis that the power law relationship may no longer be applicable as the visual scene velocity approaches zero.

One potential explanation for the absence of a clearly defined plateau is the increased relative influence of spontaneous sway arising from internal sensory and motor noise at very small stimulus velocities, as well as other sources of variability (van der Kooij & Peterka, 2011). As the visual stimulus becomes smaller, the postural sway response also becomes smaller and may therefore be more strongly affected by ongoing spontaneous sway. This could contribute to the larger variability observed at 0.01°/s.

Previous findings have shown that repeated exposure to visual motion can reduce visually evoked postural sway responses within a session, which has been interpreted as rapid habituation or adaptive changes in sensory control (Bronstein, 1986). Given this possibility, a potential session related effect was also considered, as the gain values measured in Session 2 generally appeared to be lower than those measured in Session 1. However, the 0.01°/s condition, which was the only stimulus velocity presented in both sessions, did not demonstrate such a decrease. When the two Session 1 trials were analysed separately, gain differed between Session 1a and Session 1b, indicating substantial within session variability at this velocity. Mean gain was highest in Session 2, although the large variability across all three measurements makes this difference difficult to interpret. The remaining stimulus velocities cannot be directly compared across sessions because different velocities were tested in each session. Therefore, the present data are insufficient to determine whether a session effect was present.

A potential limitation of the present study is the relatively limited amount of data available for the lowest stimulus conditions. Further trials at these velocities are required to more accurately characterize the response and determine whether gain approaches a plateau.

Furthermore, in order to evaluate a possible session effect it is essential to utilise the same stimulus velocities in both sessions.

## Conclusion

The present study investigated postural responses to extremely small visual scene velocities in order to determine whether response gain forms a plateau as visual scene velocity approaches zero. Gain was found to decrease with increasing visual scene velocity, while values between 0.03 and 0.25°/s showed a tendency to level off. However, the higher and more variable gain at 0.01°/s prevented the clear identification of a plateau. The findings indicate a potential deviation from the established gain-velocity relationship at low velocities, which is consistent with the low velocity behaviour predicted by the RFM estimator. Further measurements with a larger number of trials at the lowest velocities are required to characterise this relationship more accurately.

# References

Assländer, L., & Peterka, R. J. (2014). Sensory reweighting dynamics in human postural control. Journal of neurophysiology, 111(9), 1852-1864.

Assländer, L., Hettich, G., & Mergner, T. (2015). Visual contribution to human standing balance during support surface tilts. Human movement science, 41, 147-164.

Assländer, L., Albrecht, M., Diehl, M., Missen, K. J., Carpenter, M. G., & Streuber, S. (2023). Estimation of the visual contribution to standing balance using virtual reality. Sci Rep 13: 2594.

Assländer, L., Albrecht, M., Gruber, M., & Peterka, R. J. (2025). Is it me or the train moving? Humans resolve sensory conflicts with a nonlinear feedback mechanism in balance control. The Journal of Neuroscience, 45(29), e2303242025.

Blümle, A., Maurer, C., Schweigart, G., & Mergner, T. (2006). A cognitive intersensory interaction mechanism in human postural control. Experimental brain research, 173(3), 357-363.

Dokka, K., Kenyon, R. V., Keshner, E. A., & Kording, K. P. (2010). Self versus environment motion in postural control. PLoS computational biology, 6(2), e1000680.

Mergner, T., Siebold, C., Schweigart, G., & Becker, W. (1991). Human perception of horizontal trunk and head rotation in space during vestibular and neck stimulation. Experimental brain research, 85(2), 389-404.

Mergner, T., Maurer, C., & Peterka, R. J. (2003). A multisensory posture control model of human upright stance. Progress in brain research, 142, 189-201.

Mergner, T., Schweigart, G., Maurer, C., & Blümle, A. (2005). Human postural responses to motion of real and virtual visual environments under different support base conditions. Experimental brain research, 167(4), 535-556.

Mergner, T. (2010). A neurological view on reactive human stance control. Annual Reviews in Control, 34(2), 177-198.

Peterka, R. J., & Benolken, M. S. (1995). Role of somatosensory and vestibular cues in attenuating visually induced human postural sway. Experimental brain research, 105(1), 101-110.

Peterka, R. J. (2002). Sensorimotor integration in human postural control. Journal of neurophysiology.

Peterka, R. J., Murchison, C. F., Parrington, L., Fino, P. C., & King, L. A. (2018). Implementation of a central sensorimotor integration test for characterization of human balance control during stance. Frontiers in neurology, 9, 1045.